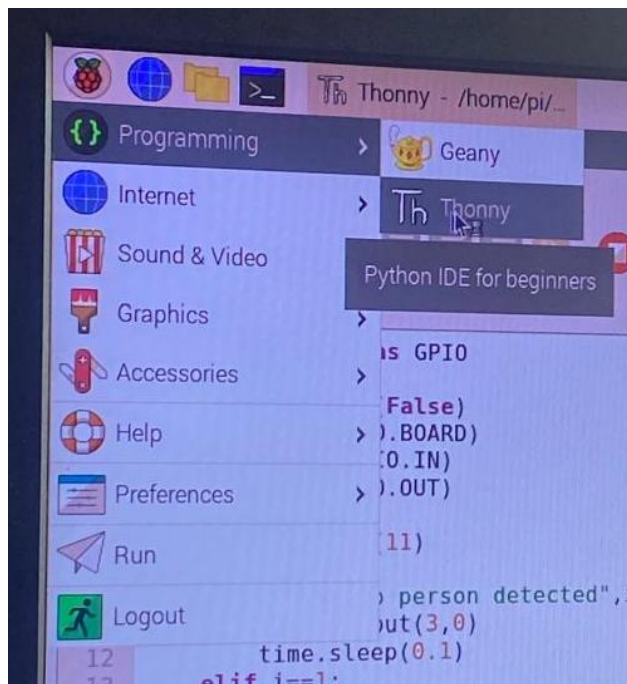


System Setup and Program Execution (Raspberry Pi)

- Connect the required peripherals to the Raspberry Pi:
 - Keyboard
 - Mouse
 - Display (Monitor)
 - Power supply (connected to electrical socket)
- After all connections are made, power ON the Raspberry Pi
- The Raspberry Pi interface (desktop screen) will open
- On the top-left corner, click on the Raspberry Pi icon
- Navigate through the following path:
 - Programming → Thonny
- The Thonny IDE window will open
- The program writing area (editor) will appear, where the code can be written and executed



General Raspberry Pi Pin Configuration

- Pin 1 and 17 → 3.3V
- Pin 2 and 4 → 5V

- Pins 6, 9, 14, 20, 25, 30, 34, 39 → Ground (GND)
-

Experiment: LED ON/OFF (not in record, given by sir)

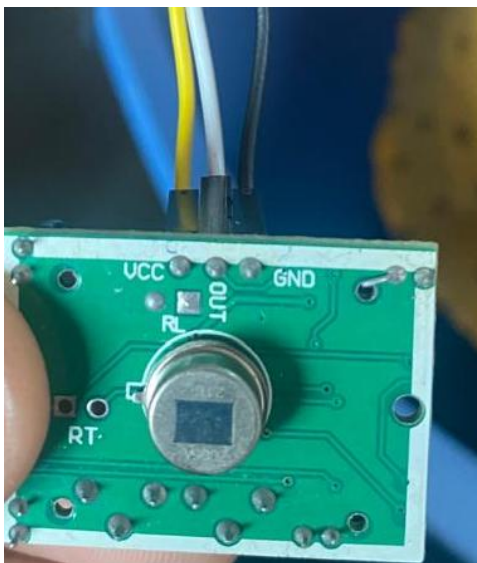
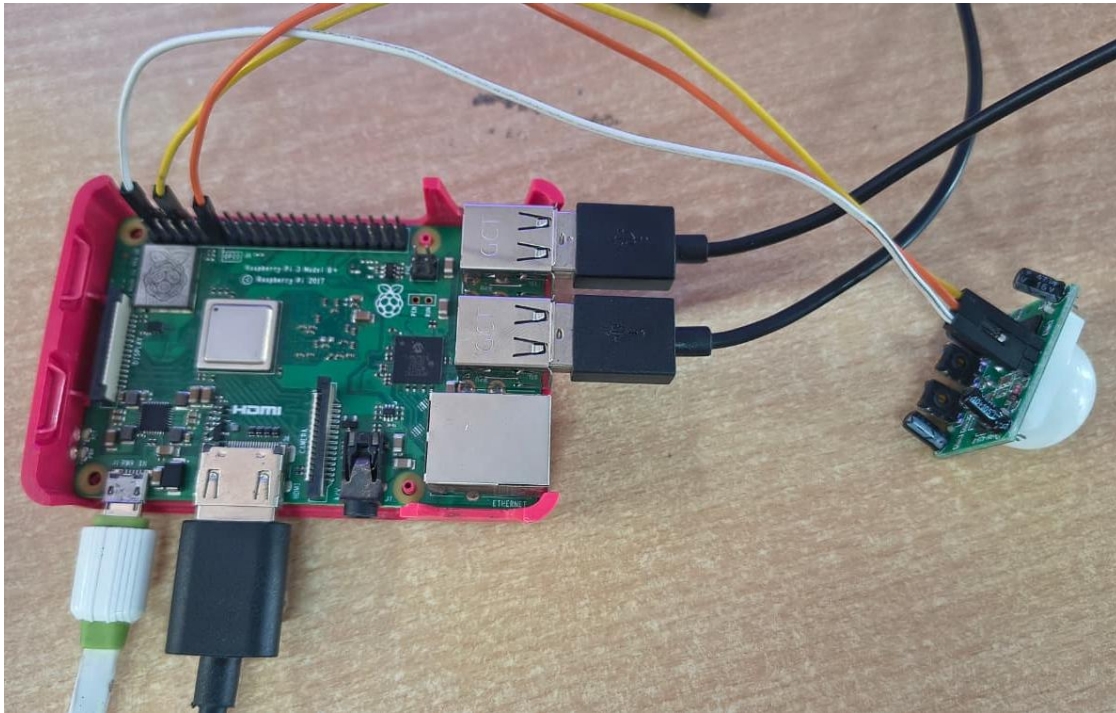


- LED is connected to **pin 10**
- Resistor is connected to **pin 6**
- Ensure **GPIO.setwarnings(False)** is included in the code

CODE:

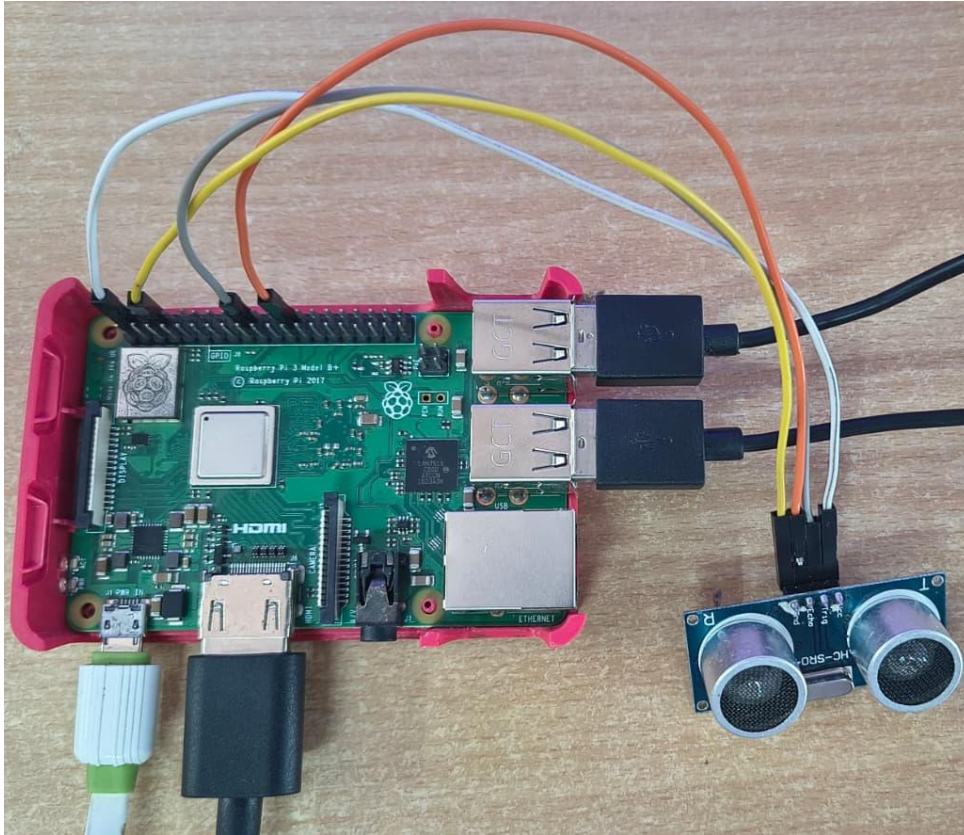
```
ledon.py x
1 import RPi.GPIO as GPIO
2 import time
3
4 GPIO.setwarnings(False)
5 GPIO.setmode(GPIO.BOARD)
6 |
7 while True:
8     GPIO.setup(10,GPIO.OUT)
9     GPIO.output(10,GPIO.HIGH)
10    time.sleep(2)
11    GPIO.output(10,GPIO.LOW)
12    time.sleep(2)
```

Experiment 6: PIR Sensor



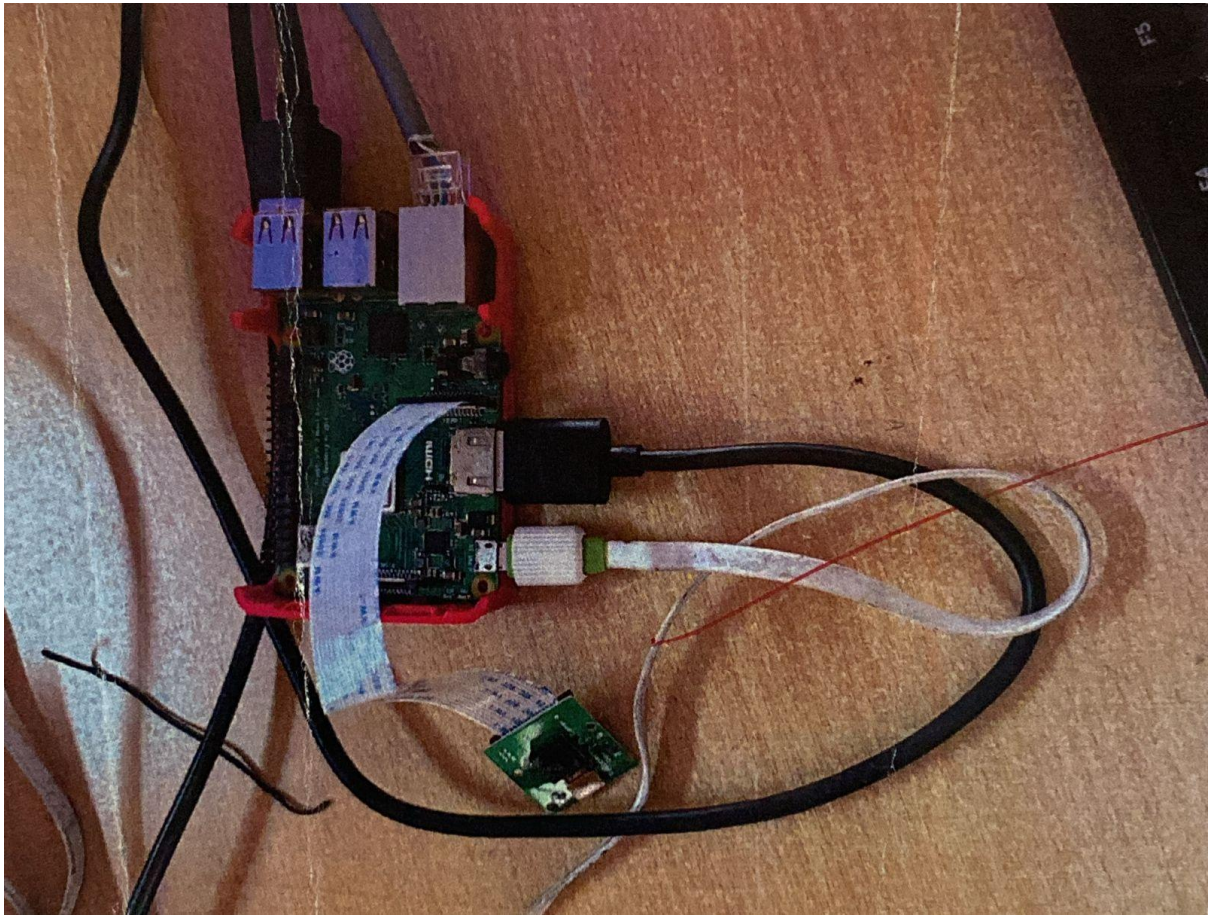
- PIR sensor has **three connections**:
 - **VCC**
 - **OUT**
 - **GND**
- **VCC** and **GND** are connected based on general configuration
- **OUT** is connected according to the code
 - In this case → **pin 3**

Experiment 8: Ultrasonic Sensor



- Ultrasonic sensor has **four connections**:
 - **VCC**
 - **Trigger**
 - **Echo**
 - **GND**
 - Connections:
 - **VCC → pin 1**
 - **Trigger → pin 18**
 - **Echo → pin 24**
 - **GND → pin 6**
-

Experiment 9: Pi Camera



- Pi Camera is **directly connected** to the Raspberry Pi